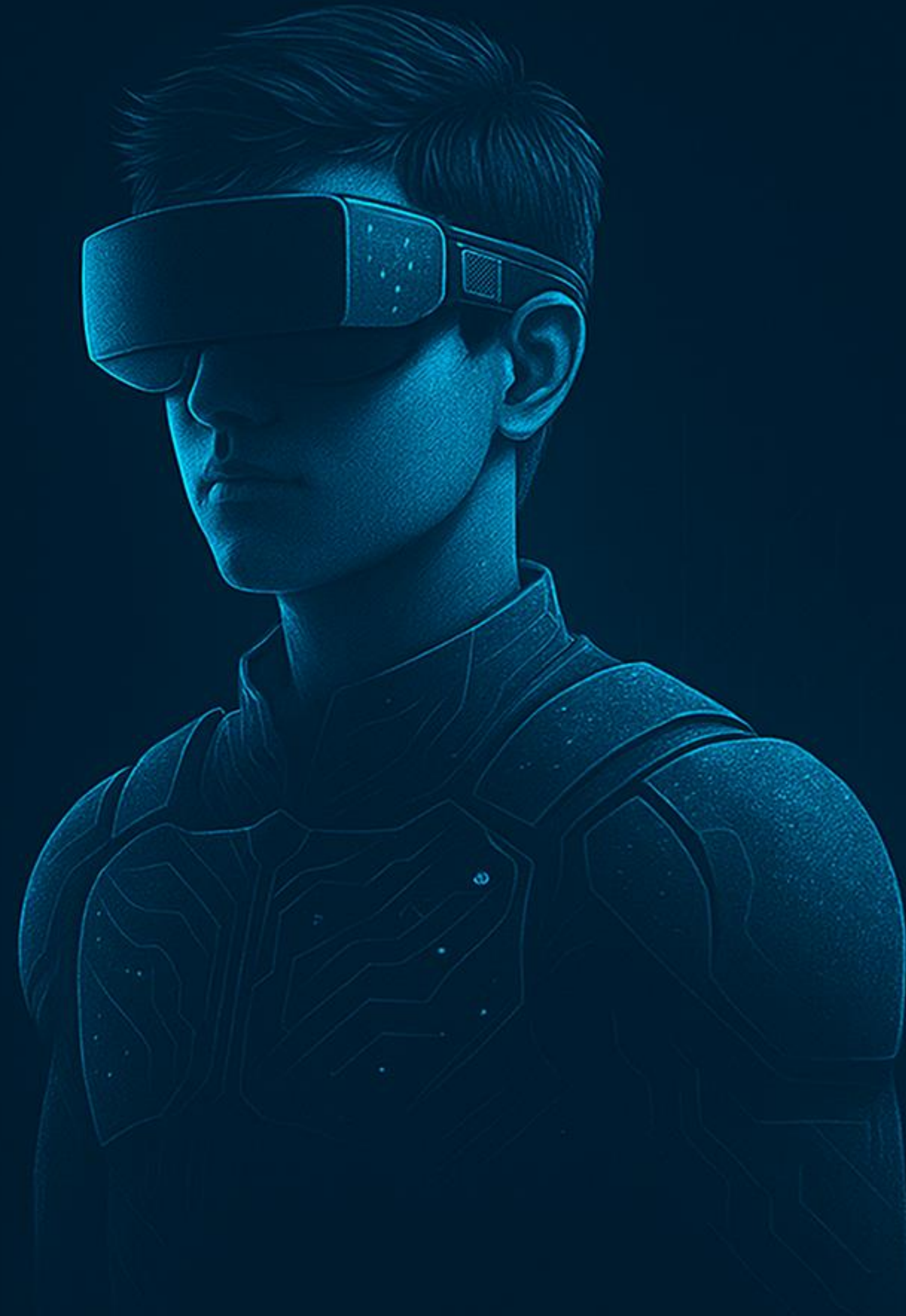
CYBERLABS

Module

4

Infrastructure & Routing/Switching-Theory



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Training structure

EASY

3X 45 MIN

LAB SIMULATION

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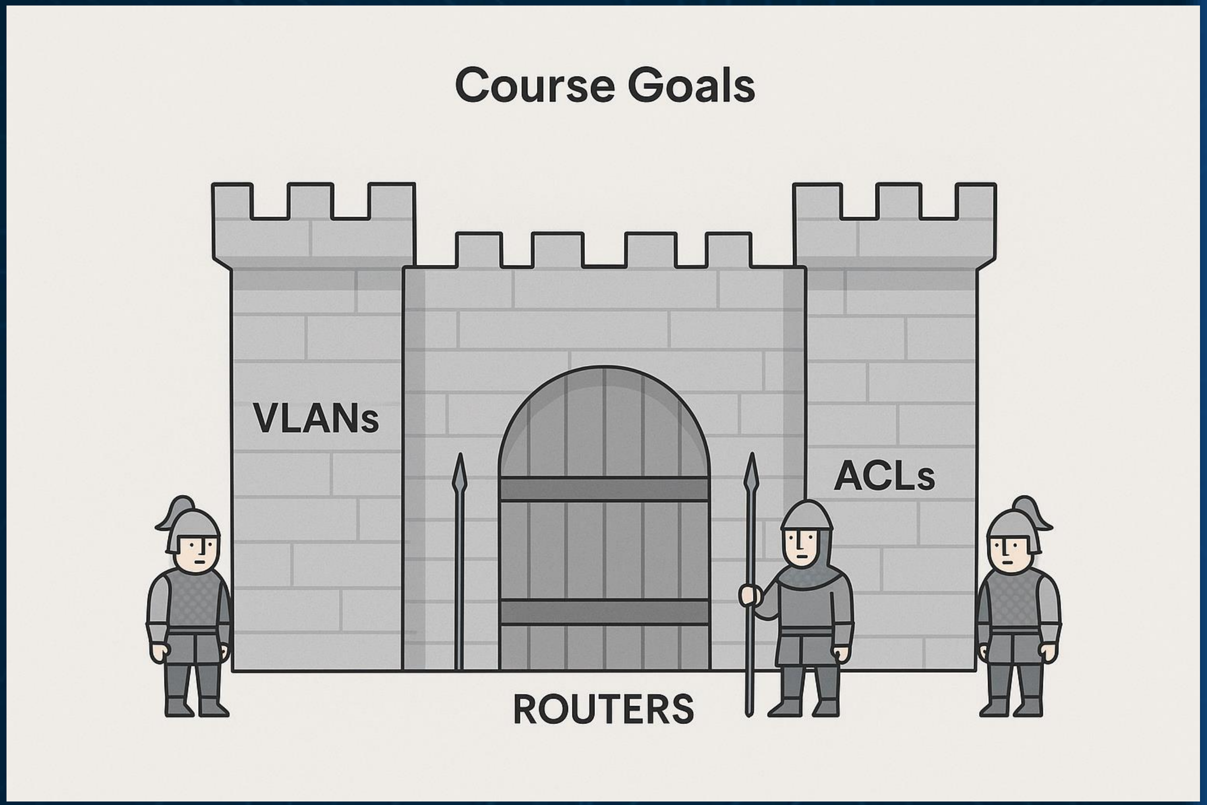
Module: 4 - Introduction to Infrastructure & Routing/Switching

- Core of secure communication networks.
- Builds on networking fundamentals.
- Topics: VLANs, routing (OSPF), ACLs, infrastructure security.

Module: 4 - Course Goals

- Key Topics:
  - Design secure, segmented networks.
  - Understand routing and switching fundamentals.
  - Apply ACLs and NAT for access control.
  - Integrate security into network infrastructure.



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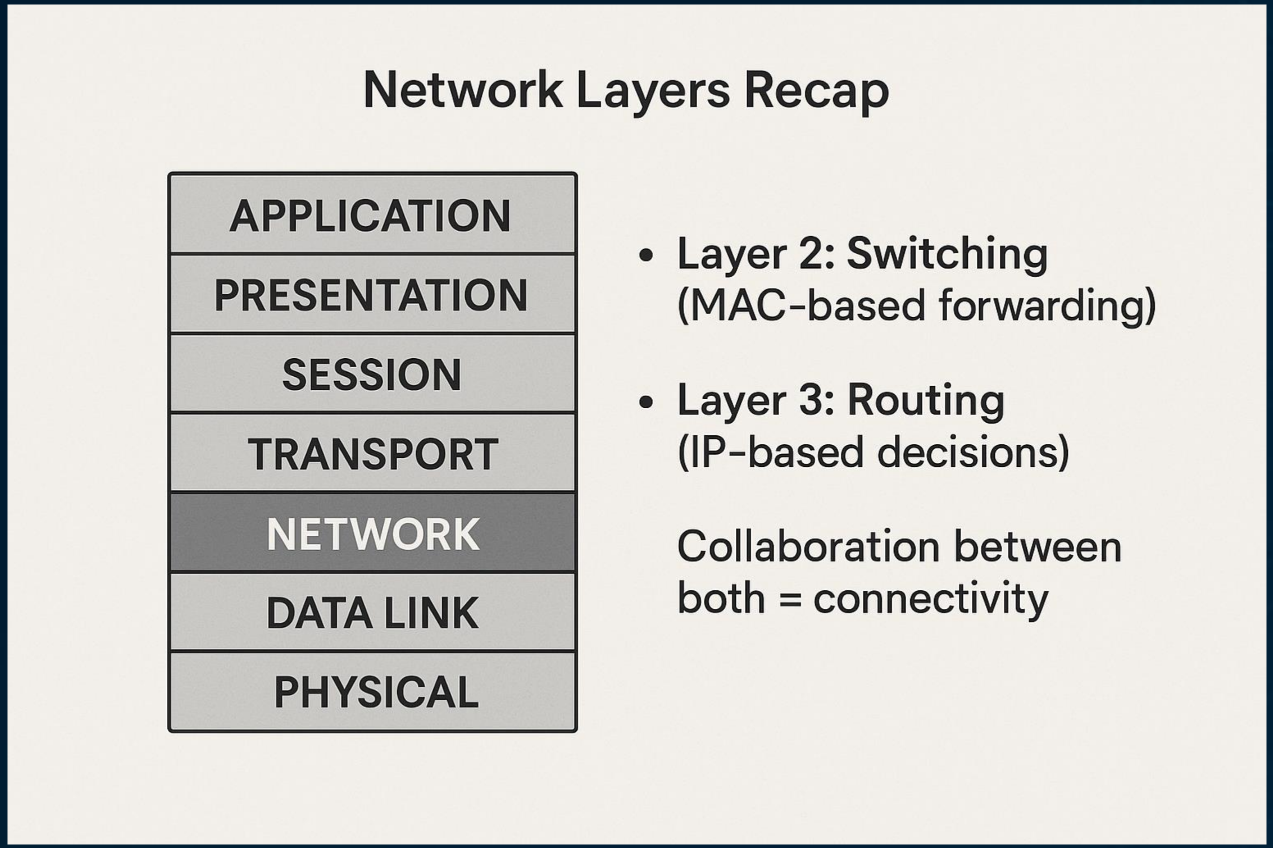
Theory Fundamentals

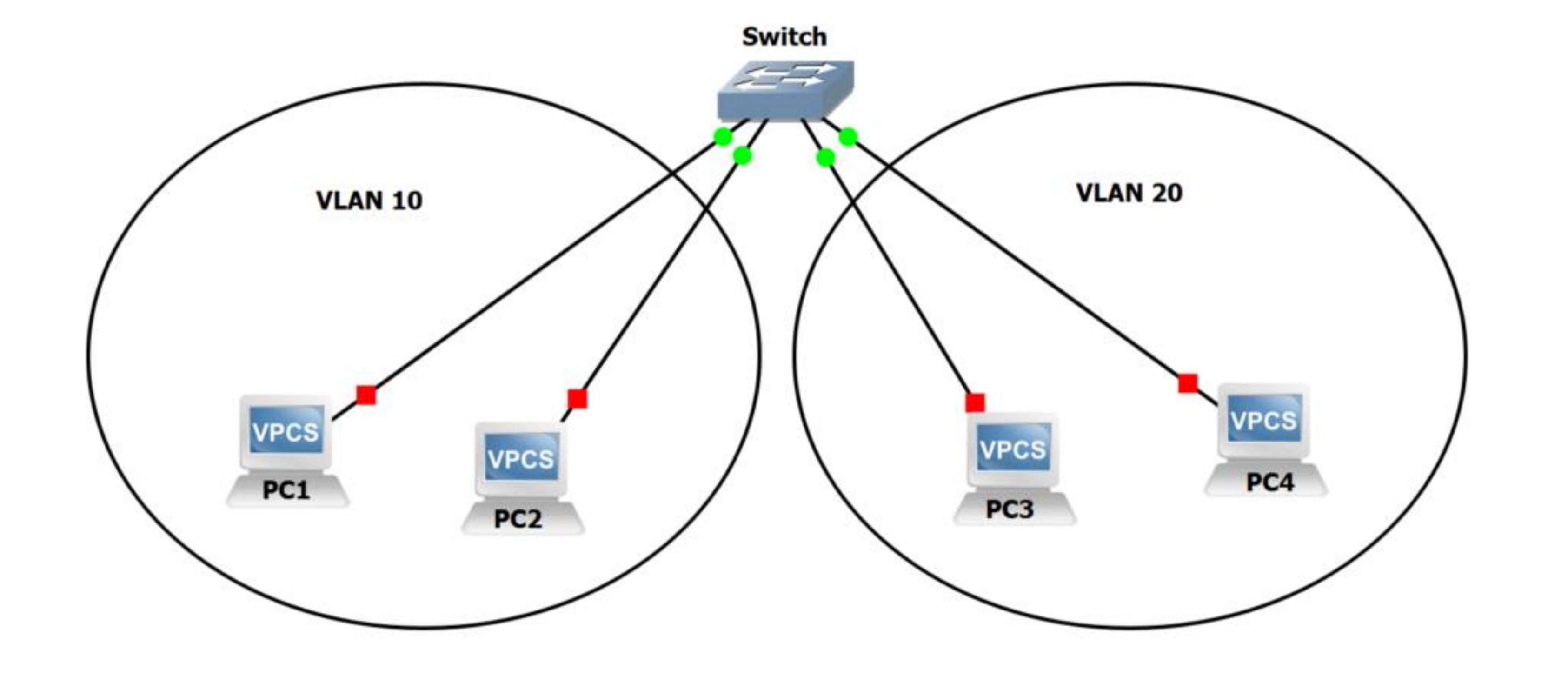
Network Layers Recap

- Layer 2: Switching (MAC-based forwarding).
- Layer 3: Routing (IP-based decisions).
- Collaboration between both = connectivity.



Switching Overview

- Operates at Layer 2 .
- Uses MAC addresses for forwarding.
- Forms a single broadcast domain.



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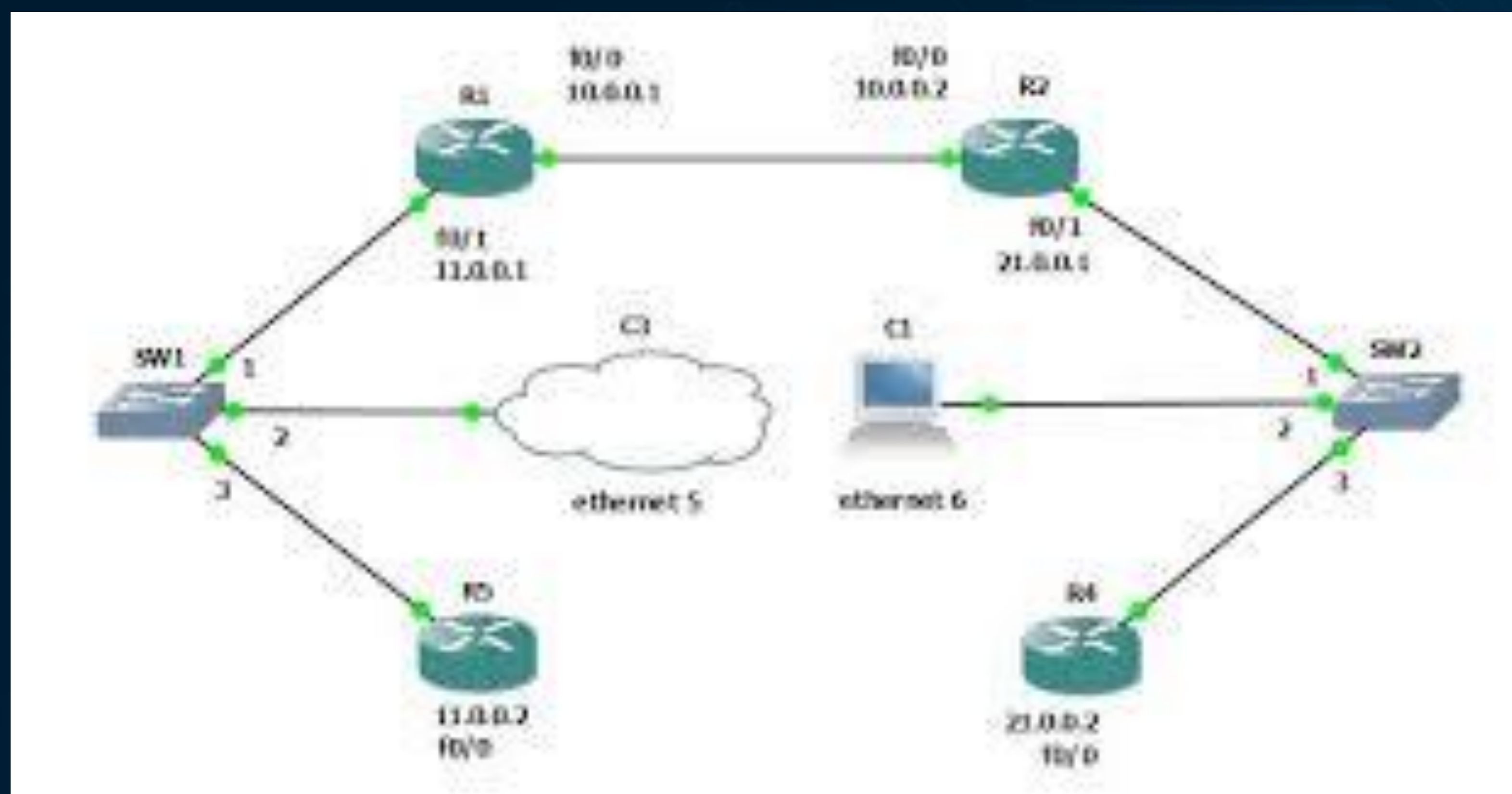
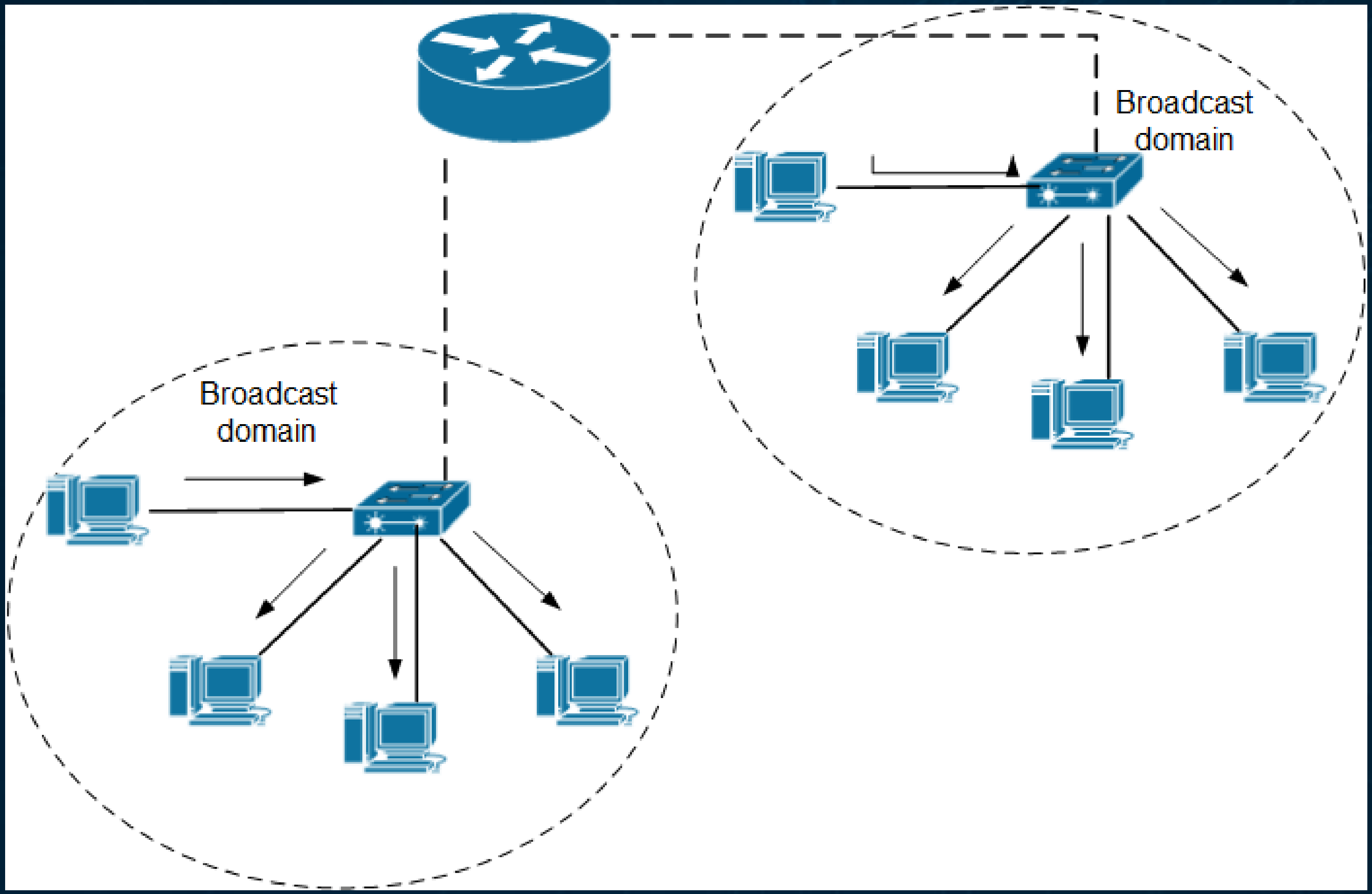
Routing Overview

- Operates at Layer 3 .
- Uses IP addressing .
- Enables communication between subnets.



Broadcast Domains & VLANs

- VLAN = Virtual LAN .
- Segments devices into isolated domains.
- Reduces broadcast traffic.
- Enhances performance & security.



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VLAN Implementation

- Each VLAN = unique ID (1–4094).
- Devices in same VLAN share broadcast domain.
- VLANs configured on managed switches.

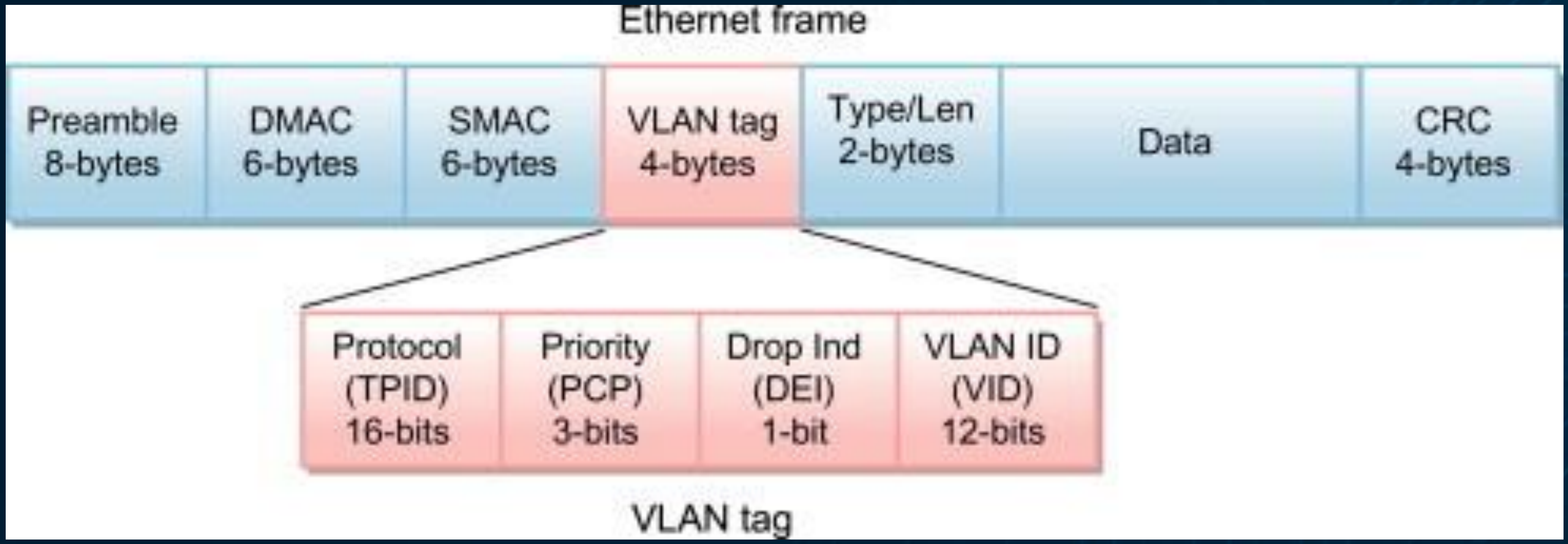
```
SW1#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
2    Accounting                       active    Fa0/5
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
2    enet  100002     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0
```

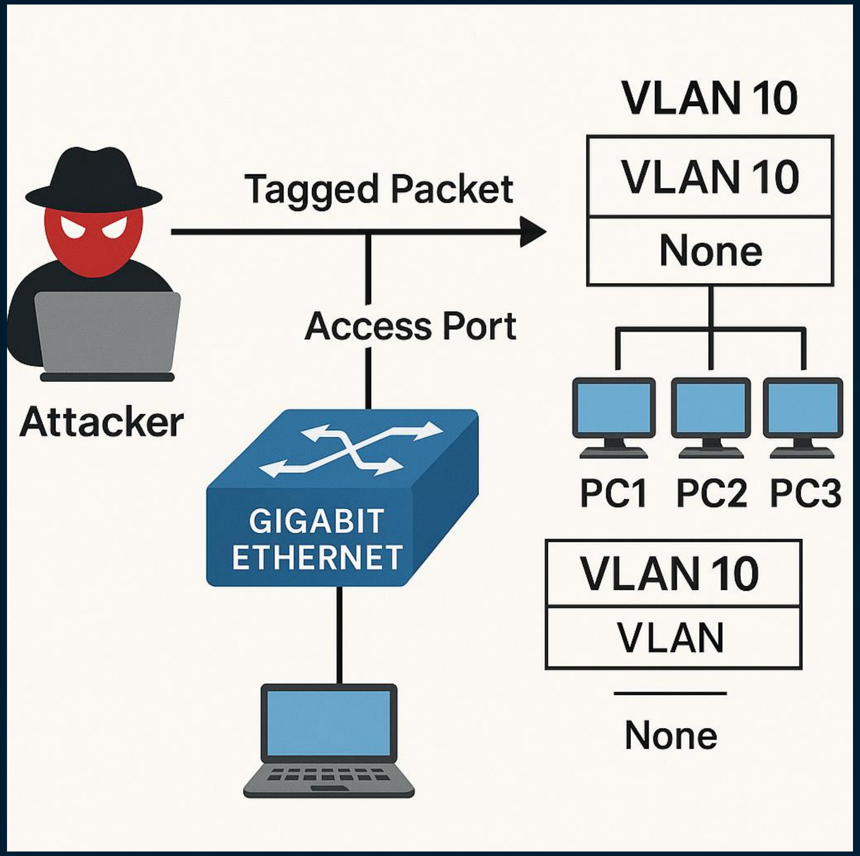
Trunking & Tagging (802.1Q)

- Access ports: one VLAN.
- Trunk ports: multiple VLANs.
- IEEE 802.1Q adds VLAN ID to Ethernet frame.



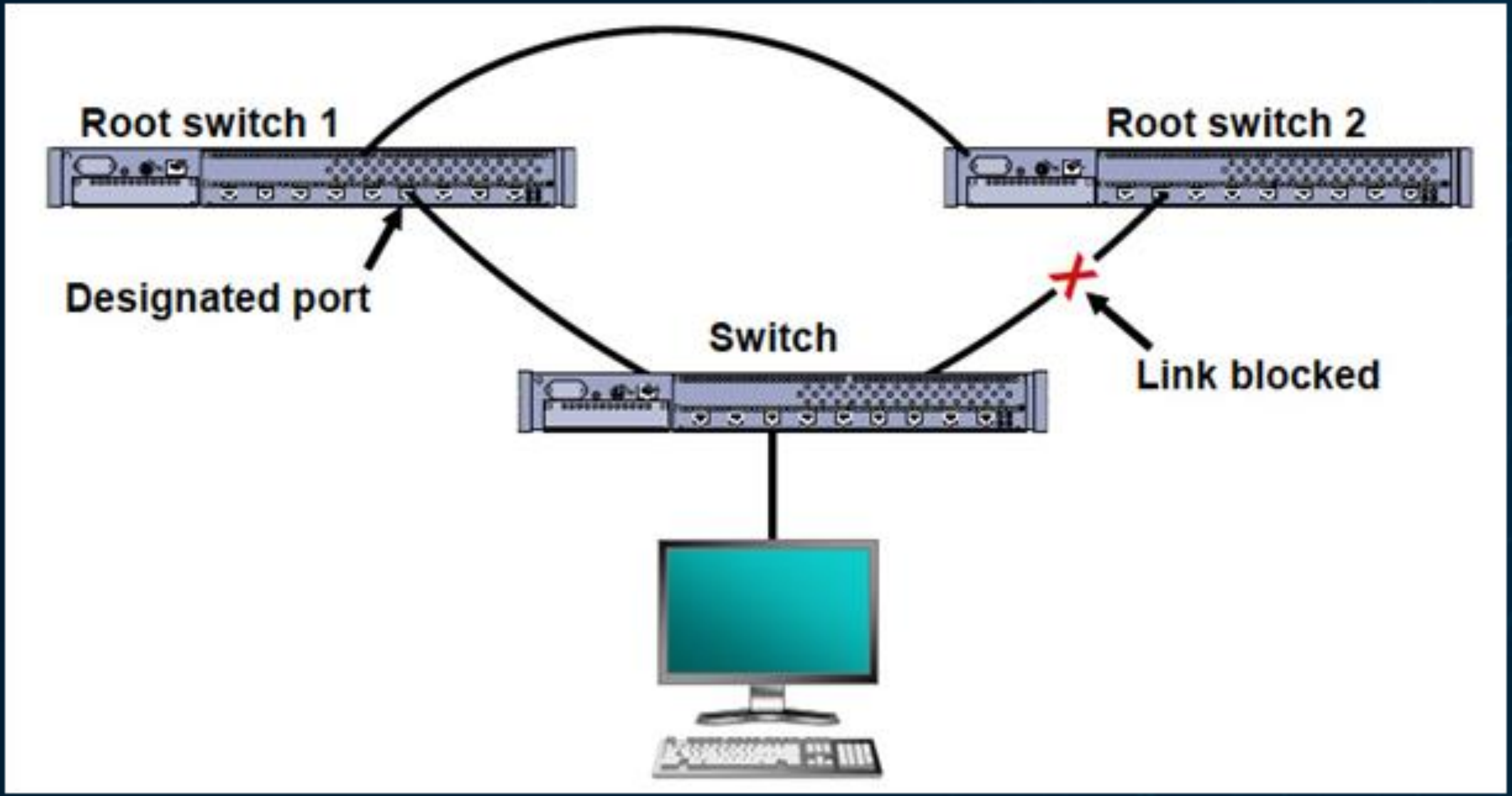
VLAN Hopping Risks

- VLAN hopping = unauthorized VLAN access.
- Exploits double-tagging or misconfigured trunks.
- Mitigate via correct native VLAN config.



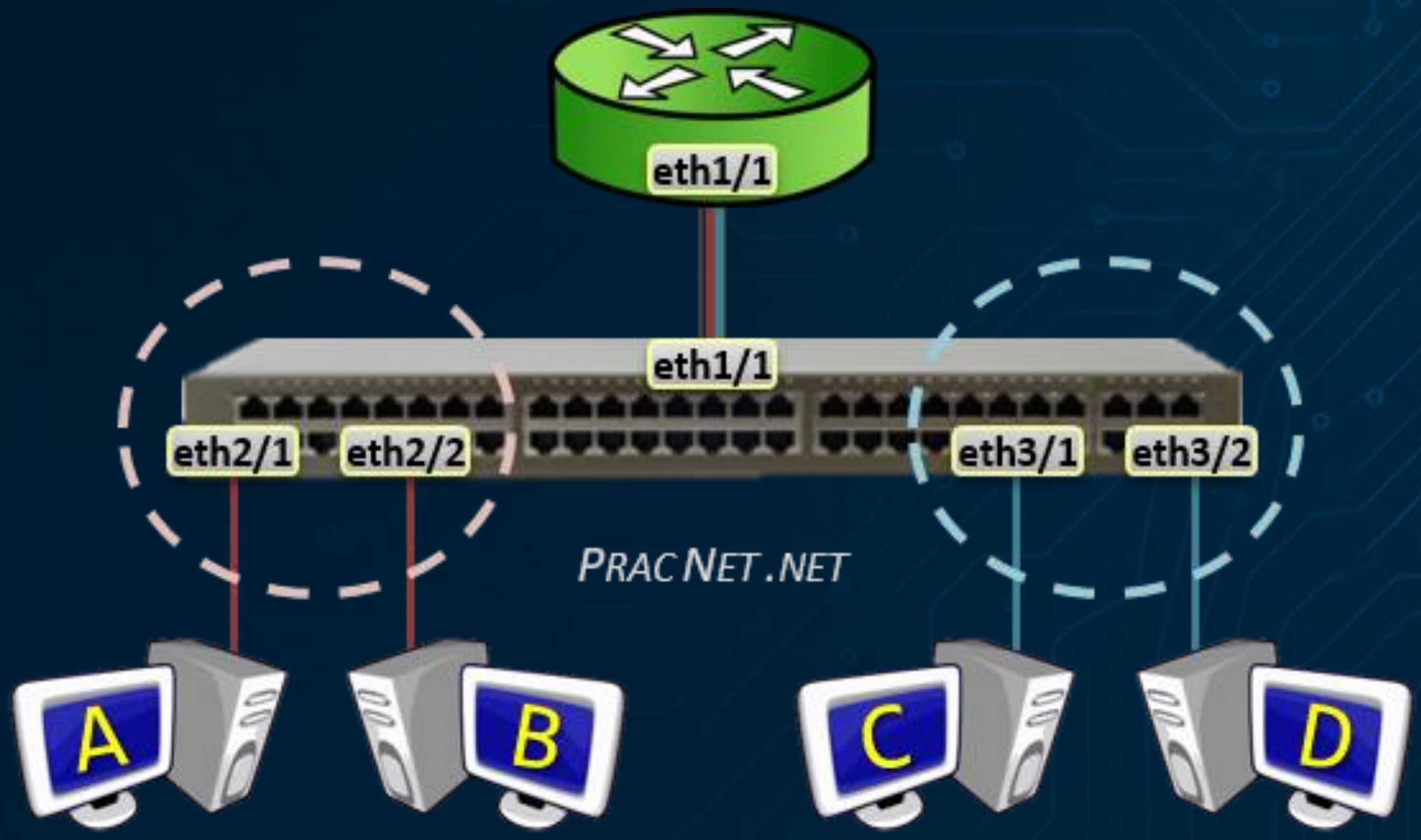
Spanning Tree Protocol (STP)

- Prevents network loops.
- Selects root bridge & blocks redundant paths.
- BPDU Guard protects STP from attacks.



Layer 3 Overview

- Router (or L3 switch) connects VLANs.
- Each VLAN = distinct subnet.
- Routing enables inter-VLAN communication.



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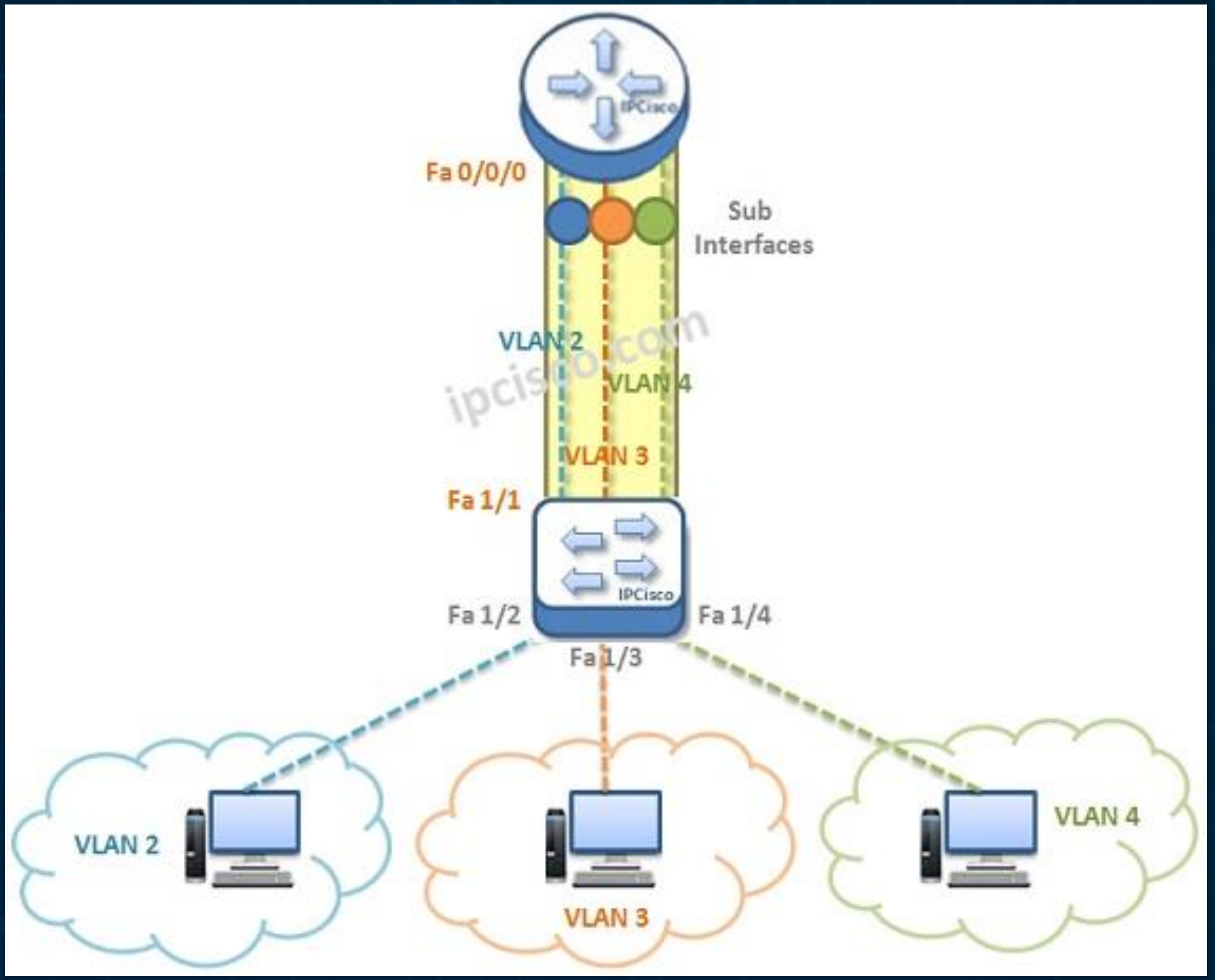
Inter-VLAN Routing

- Uses router subinterfaces per VLAN.
- Each subinterface tagged for VLAN.
- Common in smaller networks.



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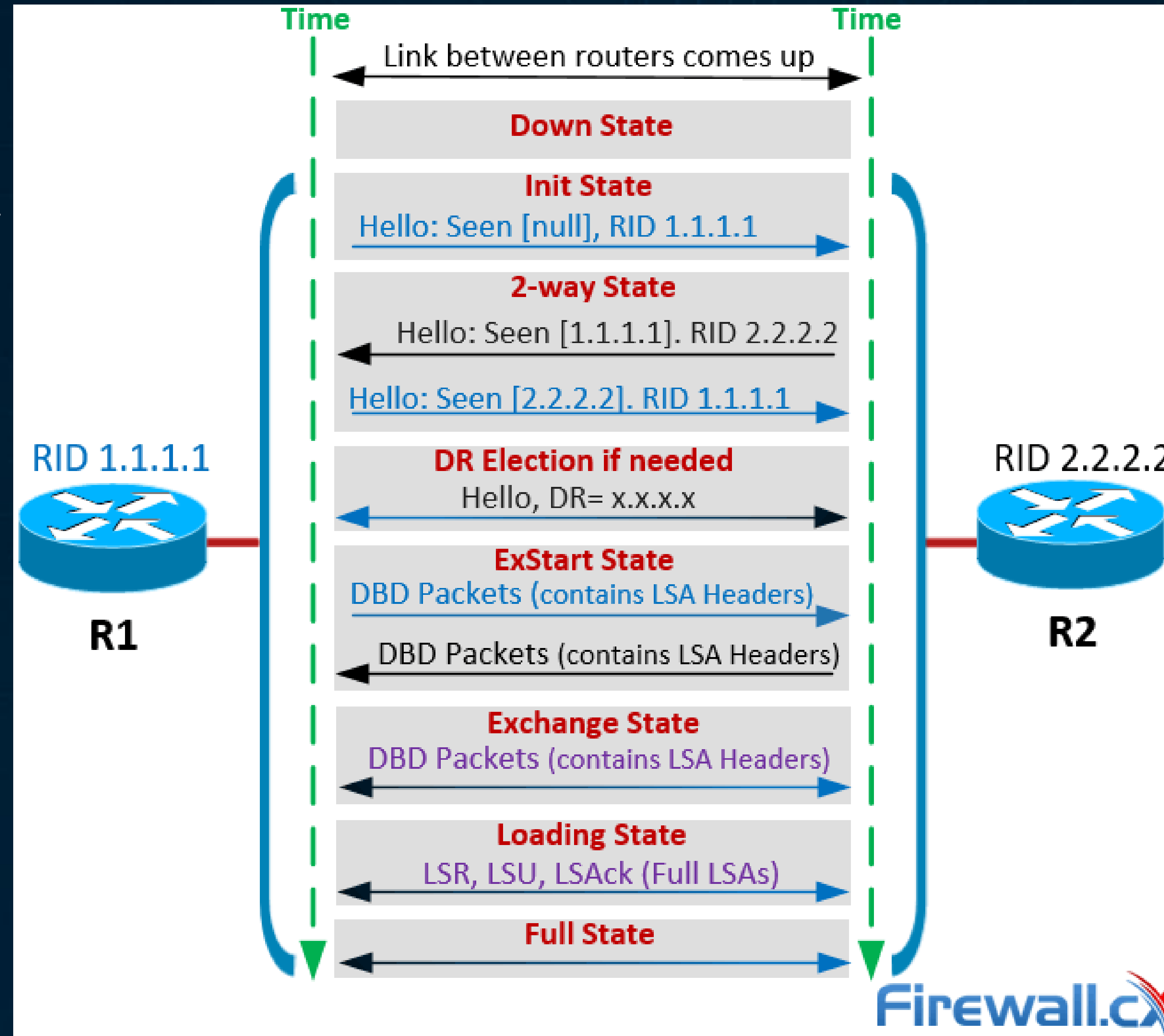
Static Routing

- Manually configured paths.
- Syntax: `ip route 192.168.10.0 255.255.255.0 192.168.1.1`
- Best for small, stable environments.

```
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.0.1      192.168.0.5     20
       10.128.4.0  255.255.255.192     10.128.64.10      192.168.0.5     21
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    306
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    306
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    306
      192.168.0.0    255.255.255.0         On-link       192.168.0.5    276
      192.168.0.5  255.255.255.255         On-link       192.168.0.5    276
    192.168.0.255  255.255.255.255         On-link       192.168.0.5    276
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    306
        224.0.0.0        240.0.0.0         On-link       192.168.0.5    276
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    306
  255.255.255.255  255.255.255.255         On-link       192.168.0.5    276
===========================================================================
```

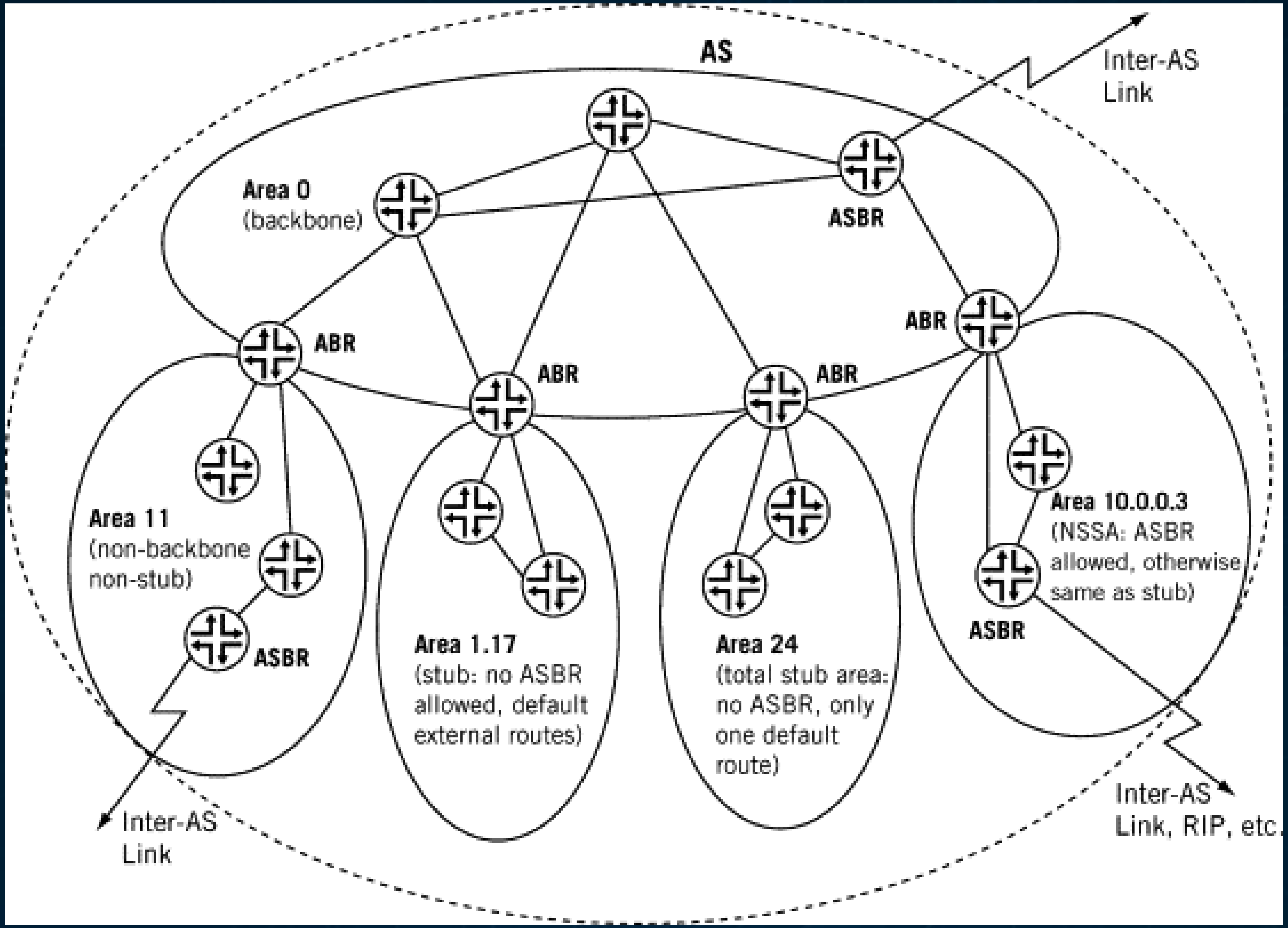
Dynamic Routing Basics

- Routers share routes automatically.
- Updates triggered by topology changes.
- Examples: OSPF, EIGRP, BGP.



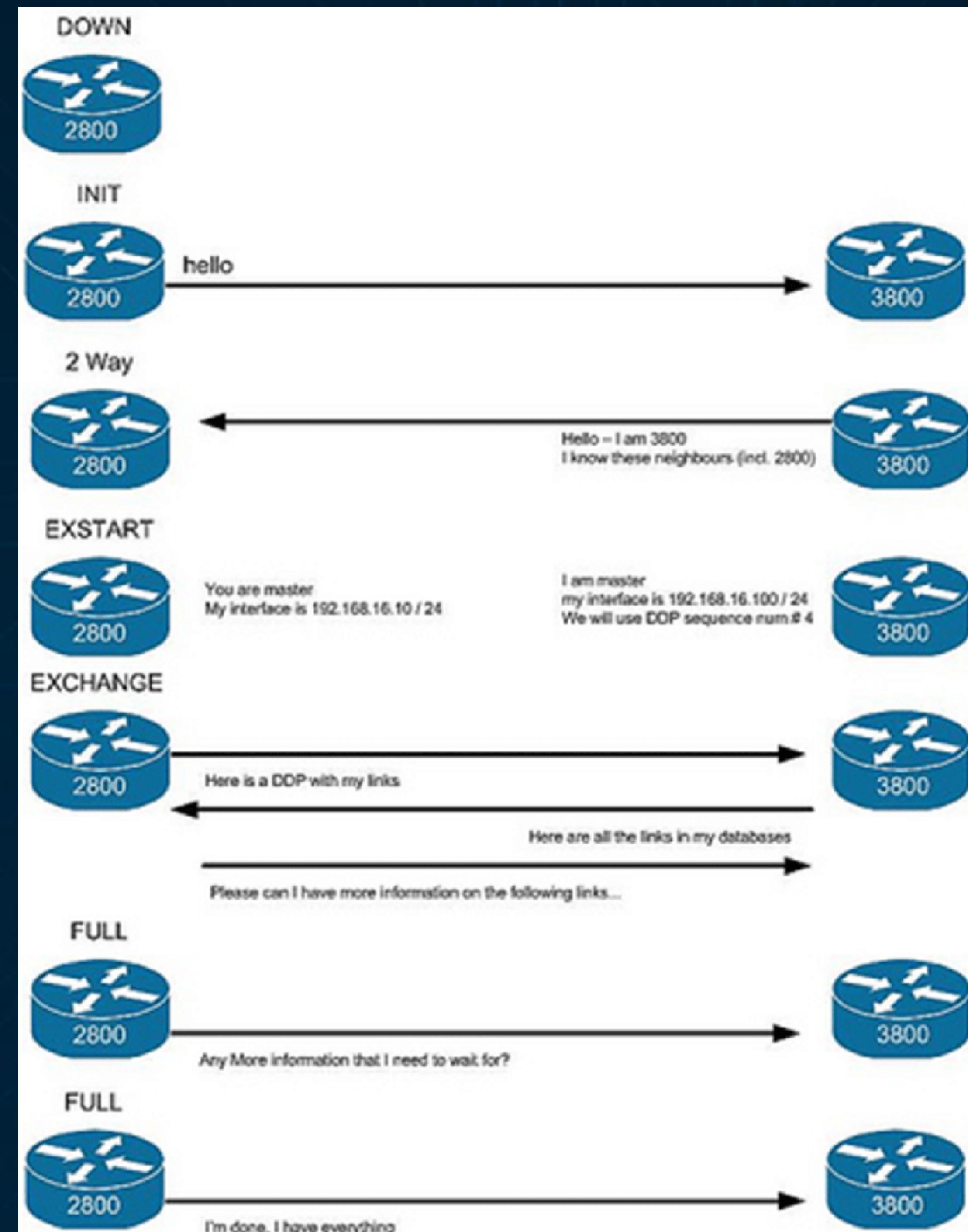
OSPF Overview

- Link-state dynamic protocol.
- Calculates shortest path (Dijkstra).
- Organized into areas (Area 0 = backbone).



OSPF Neighbor Discovery

- Routers send periodic Hello packets.
- Establish adjacency before exchanging LSAs.
- Must match area ID, timers, authentication.



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OSPF Metrics

- Cost = 100 Mbps / Interface Bandwidth.
- Lower cost = preferred route.
- Can adjust manually.

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Route Summarization

- Reduces routing table entries.
- Hides internal topology.
- Improves stability.

```
R2#show ip bgp
BGP table version is 5, local router ID is 10.10.20.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.10.0.0/16     23.23.23.3               0             0 3 i
*> 10.10.20.0/24    0.0.0.0                  0         32768 i
R2#
```

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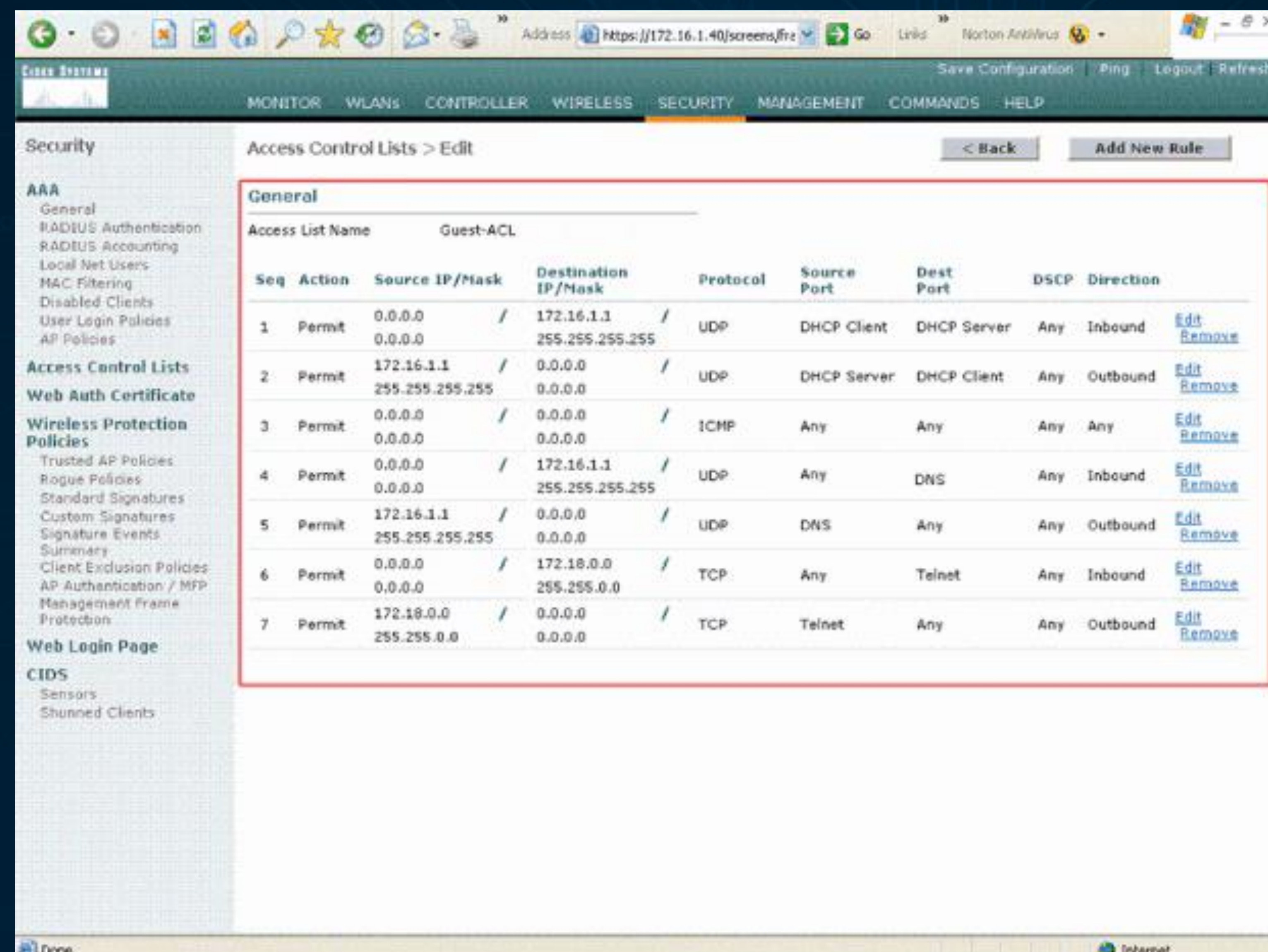
Access Control Lists (ACLs)

- Traffic filters on routers/switches.
- Based on IP, port, or protocol.
- Enforces security policies.



ACL Types

Type	Filters On	Example
Standard	Source IP	access-list 10 permit 192.168.10.0
Extended	Src/Dst + Port	access-list 110 permit tcp any any eq 80

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ACL Order & Implicit Deny

- Evaluated top-down.
- First match wins.
- Ends with implicit deny all .
- Some vendor exceptions:
  - Palo Alto uses zone-based policies with application awareness before port matching
  - Fortinet offers both policy-based and route-based firewall modes
  - Linux firewalls require explicit default policy setting (critical security decision)
  - Windows Firewall has asymmetric defaults (block inbound, allow outbound)

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NAT & PAT Concepts

- NAT hides internal addresses.
- PAT uses port numbers for multiplexing.
- Increases address efficiency.

- NAT: Network Address Translation. Exit IP appears to be external IP (192.168.1.1 -> 33.44.55.3)
- PAT: Port Address Translation. Port is forwarded to internal address: 33.44.55.3:80 -> 192.168.1.4:8080 and 33.44.55.3:443 -> 192.168.1.5:443

Infrastructure Hardening

- Disable unused interfaces.
- Use SSH , not Telnet.
- Implement SNMPv3 .
- Enforce strong passwords & AAA.

Time & Logging

- Enable Syslog for change monitoring.
- Sync clocks using NTP .
- Retain logs for compliance.

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Layer 2 Threats

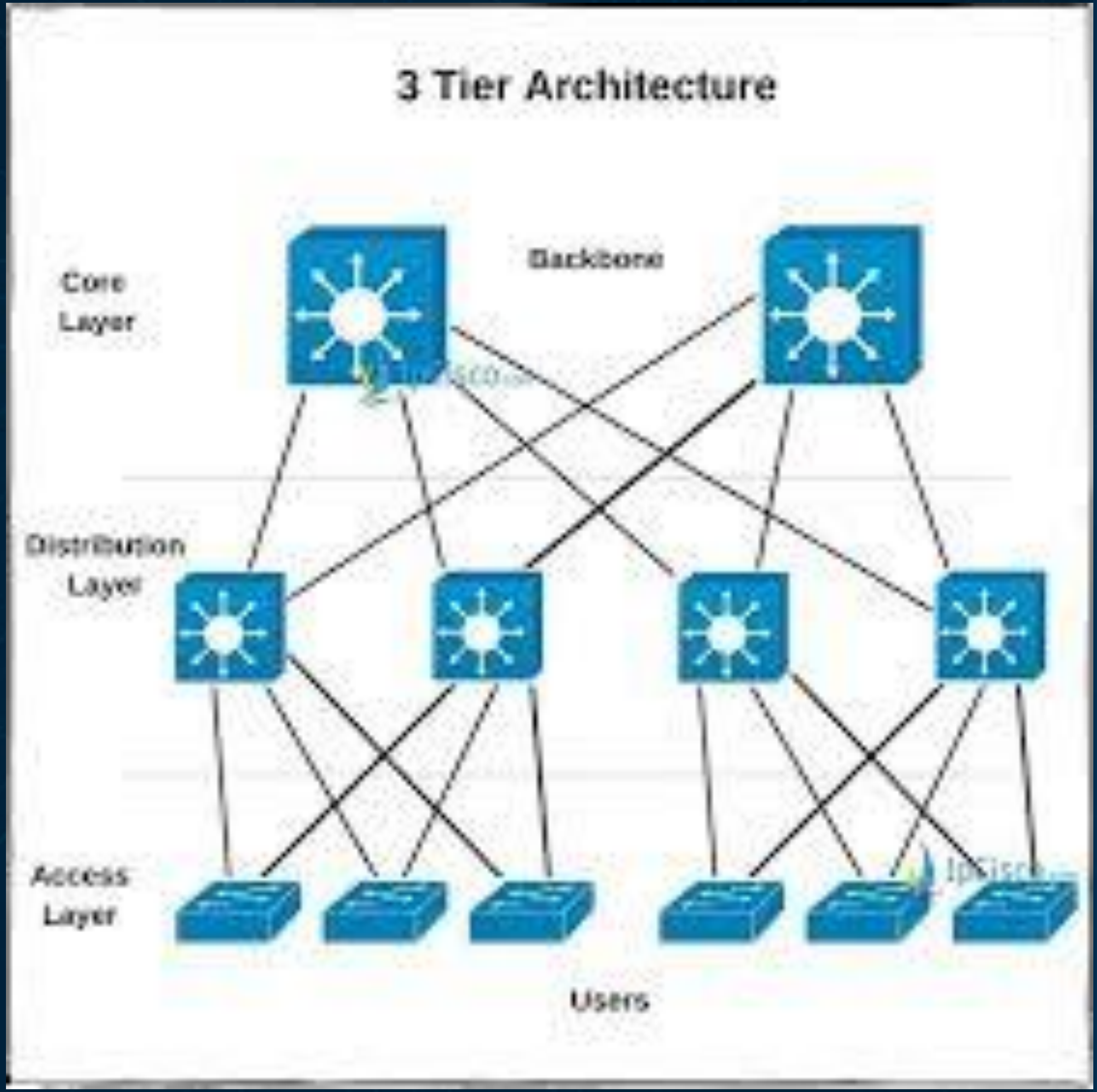
Threat	Description
ARP Spoofing	Fake MAC-IP mapping
VLAN Hopping	Unauthorized VLAN access
STP Attack	Fake root bridge

Layer 2 Mitigation Techniques

- Enable Port Security .
- Use DHCP Snooping .
- Lock MAC addresses.
- Disable unused ports.

Network Design Principles

- Core: High-speed transport.
- Distribution: Routing & policy.
- Access: End-user connectivity.

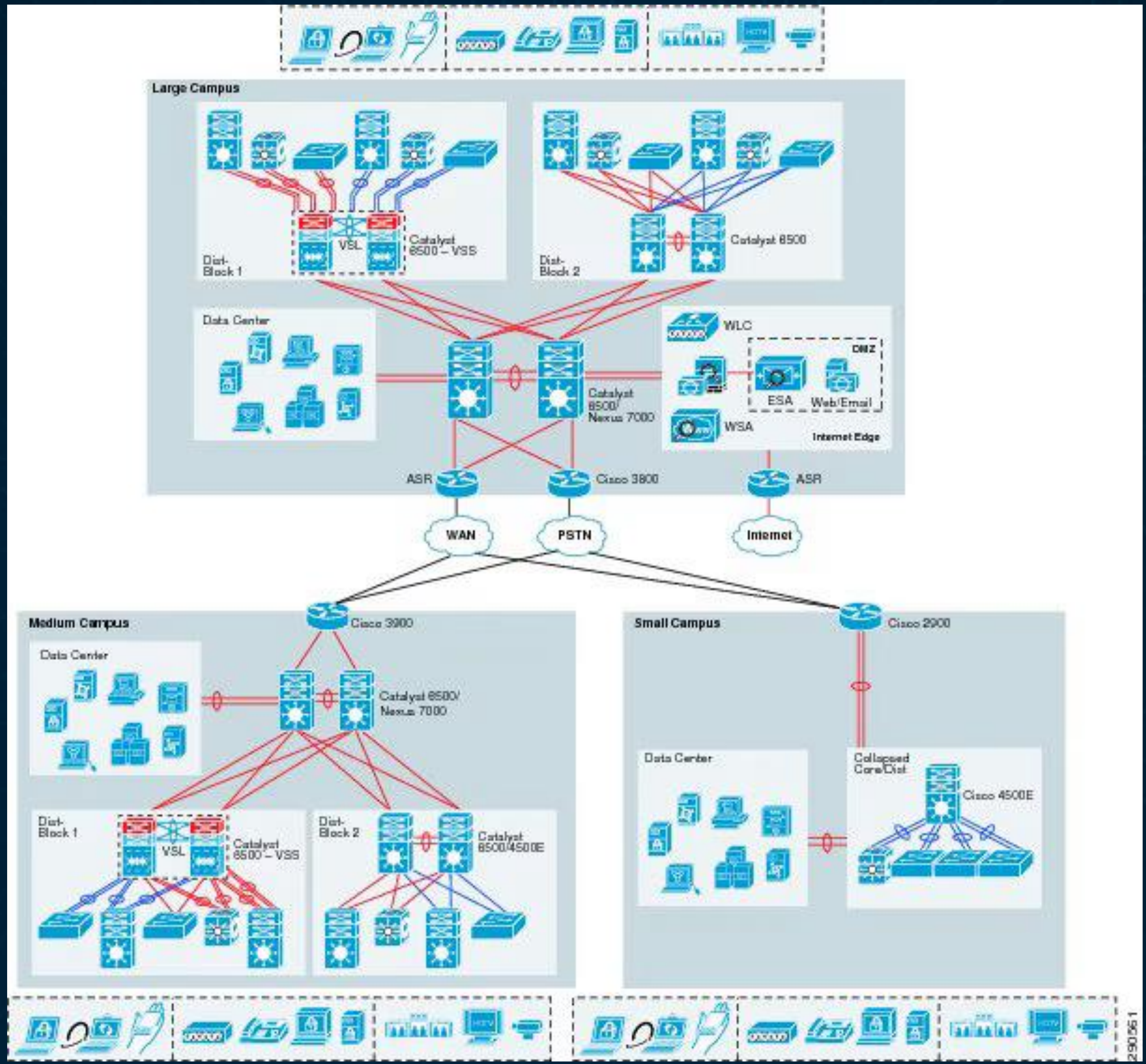


Redundancy Concepts

- HSRP / VRRP: Router redundancy.
- STP: Prevents loops.
- EtherChannel: Link aggregation.

Documentation Standards

- Maintain topology diagrams.
- Record VLANs, ACLs, IPs.
- Version control configs.



Monitoring Infrastructure

- Centralize router/switch logs.
- Monitor configuration changes.
- Correlate with SOC alerts.

Secure Management Practices

- Restrict admin IPs.
- Separate management VLAN.
- Use MFA for privileged access.

Change Management

- Follow CAB approval process.
- Document config deltas.
- Rollback plans required.

- CAB: Change Advisory Board. See ITIL definition. Meeting where the impact is made clear to stakeholders and decision is made to go ahead with the change or to defer until a later time

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Compliance Alignment

- ISO 27001 A.13.1: Network security management.
- GDPR Art. 32: Integrity & availability.
- SOC 2 CC6.6: Change and configuration control.

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Review Questions

1. What separates VLANs logically?
2. How does OSPF differ from static routing?
3. Why does ACL order matter?
4. What risks exist in trunk misconfiguration?

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Summary

- VLANs = Segmentation.
- OSPF = Intelligent routing.
- ACLs = Policy enforcement.
- Security = Infrastructure hardening.

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Thank you